

第四章 向量组与线性方程组解的结构(1)

知识点 n 维向量的加法、 n 维向量的数乘、向量的线性组合、向量的线性表示、向量组的等价、向量组的线性相关性

课后练习

1. 判断题:

1) 若向量组 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性相关, 则 α_1 可由 $\alpha_2, \dots, \alpha_n$ 线性表示. ($\times$)

2) 两个等价的向量组所含的向量的个数相等. ($\times$)

3) 若矩阵 $AB=C$, 则 C 的行向量组能由 B 的行向量组线性表示. ($\checkmark$)

4) 设矩阵 A 和 B 等价, 则 A 的列向量组与 B 的列向量组等价. ($\times$)

5) 若两个向量组等价, 其中一个线性相关, 则另一个也线性相关. ($\times$)

2. 填空题:

1) 已知 $\alpha = (3, 5, 7, 9)^T, \beta = (-1, 5, 2, 0)^T$, x 满足 $2\alpha + 3x = \beta$, 则 $x = (-\frac{7}{3}, -\frac{5}{3}, -4, -6)^T$

2) 当 $k = -8$ 时, 向量 $\beta = (1, k, 5)^T$ 能由 $\alpha_1 = (1, -3, 2)^T, \alpha_2 = (2, -1, 1)^T$ 线性表出.

3) 若向量组 $\alpha_1 = \begin{pmatrix} a \\ 1 \\ 1 \end{pmatrix}, \alpha_2 = \begin{pmatrix} 1 \\ a \\ -1 \end{pmatrix}, \alpha_3 = \begin{pmatrix} 1 \\ -1 \\ a \end{pmatrix}$ 线性相关, 则 $a = -1 \text{ 或 } 2$

4) 若向量组 $\alpha_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \alpha_2 = \begin{pmatrix} k \\ 3 \\ 0 \end{pmatrix}, \alpha_3 = \begin{pmatrix} -1 \\ 4 \\ k \end{pmatrix}$ 线性无关, 则 k 满足 $k \neq -3$

3. 填写线性相关或线性无关:

- 1) 单独一个零向量是 线性相关;
- 2) 任何一个非零向量是 线性无关;
- 3) 两个成比例向量是 线性相关;
- 4) 含有零向量的向量组必 线性相关;
- 5) 一个向量组的部分组线性相关, 则该向量组必 线性相关;
- 6) 一个向量组线性无关, 则该向量组的任何部分组必 线性无关;
- 7) 一个向量组线性无关, 则延长分量后所得向量组必 线性无关;
- 8) 一个向量组线性相关, 则缩短分量后所得向量组必 线性相关;
- 9) 当 $m > n$ 时, m 个 n 维列向量必 线性相关.

4. $\alpha_1 = \begin{pmatrix} 0 \\ 0 \\ c_1 \end{pmatrix}, \alpha_2 = \begin{pmatrix} 0 \\ 1 \\ c_2 \end{pmatrix}, \alpha_3 = \begin{pmatrix} 1 \\ -1 \\ c_3 \end{pmatrix}, \alpha_4 = \begin{pmatrix} -1 \\ 1 \\ c_4 \end{pmatrix}$, c_1, c_2, c_3, c_4 为任意常数, 下列向量组线性相

关的是 (C) A. $\alpha_1, \alpha_2, \alpha_3$ B. $\alpha_1, \alpha_2, \alpha_4$ C. $\alpha_1, \alpha_3, \alpha_4$ D. $\alpha_2, \alpha_3, \alpha_4$

5. 设 $\alpha_1, \alpha_2, \alpha_3$ 为 n 维列向量, $\beta_1 = \alpha_1 + \alpha_2, \beta_2 = \alpha_2 + \alpha_3, \beta_3 = \alpha_1 + \alpha_3$, 求一个矩阵 C 使

得 $(\beta_1, \beta_2, \beta_3) = (\alpha_1, \alpha_2, \alpha_3)C$.

$$C = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

6. 设 $\alpha_1, \alpha_2, \alpha_3$ 为 n 维行向量, $\beta_1 = \alpha_1 + \alpha_2, \beta_2 = \alpha_2 + \alpha_3, \beta_3 = \alpha_1 + \alpha_3$, 求一个矩阵 C 使

$$\begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = C \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

7. 已知向量组 $\alpha_1 = \begin{pmatrix} 1 \\ 2 \\ -3 \\ 1 \end{pmatrix}, \alpha_2 = \begin{pmatrix} 5 \\ -5 \\ t \\ 11 \end{pmatrix}, \alpha_3 = \begin{pmatrix} 2 \\ -1 \\ 3 \\ t \end{pmatrix}, \alpha_4 = \begin{pmatrix} 1 \\ -3 \\ 6 \\ 3 \end{pmatrix}$, 问:

(1) 当 t 为何值时, 向量组 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 线性相关? $t = 12 \text{ or } 4$

(2) 当 t 为何值时, 向量组 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 线性无关? $t \neq 4 \text{ 且 } t \neq 12$

(3) 当 t 为何值时, α_3 能由 $\alpha_1, \alpha_2, \alpha_4$ 线性表出? $t = 4$

$$\begin{pmatrix} \alpha_1 & \alpha_4 & \alpha_2 & \alpha_3 \\ \begin{pmatrix} 1 & 1 & 5 & 2 \\ 2 & -3 & -5 & -1 \\ -3 & 6 & t & 3 \\ 1 & 3 & 11 & t \end{pmatrix} \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 5 & 2 \\ 0 & -5 & -15 & -5 \\ 0 & 9 & 15+t & 9 \\ 0 & 2 & 6 & t-2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 5 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & t-12 & 0 \\ 0 & 0 & 0 & t-4 \end{pmatrix}$$

$$(1) |(\alpha_1 \alpha_2 \alpha_3 \alpha_4)| = 0$$

$$(3) R(\alpha_1 \alpha_2 \alpha_4) = R(\alpha_1 \alpha_2 \alpha_3 \alpha_4)$$

$$R(\alpha_1 \alpha_2 \alpha_3 \alpha_4) < 4$$

8. 设向量组 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, 判断向量组 $\beta_1, \beta_2, \beta_3$ 的线性相关性:

1) $\beta_1 = \alpha_1 + \alpha_2, \beta_2 = 2\alpha_2 + 3\alpha_3, \beta_3 = 5\alpha_1 + 3\alpha_2;$

2) $\beta_1 = 2\alpha_1 + 2\alpha_2 + 4\alpha_3, \beta_2 = 3\alpha_1 + \alpha_2 + 3\alpha_3, \beta_3 = \alpha_1 + 2\alpha_2 + 3\alpha_3;$

3) $\beta_1 = \alpha_1 - \alpha_2, \beta_2 = 2\alpha_2 + \alpha_3, \beta_3 = \alpha_1 + \alpha_2 + \alpha_3$. 线性相关 $\beta_3 = \beta_2 + \beta_1$

(1) 设 $\lambda_1 \beta_1 + \lambda_2 \beta_2 + \lambda_3 \beta_3 = 0$

$$\Rightarrow (\lambda_1 + 5\lambda_3) \alpha_1 + (\lambda_1 + 2\lambda_2 + 3\lambda_3) \alpha_2 + 3\lambda_2 \alpha_3 = 0$$

$$\Leftrightarrow \begin{cases} \lambda_1 + 5\lambda_3 = 0 \\ \lambda_1 + 2\lambda_2 + 3\lambda_3 = 0 \\ 3\lambda_2 = 0 \end{cases} \Rightarrow \lambda_1 = \lambda_2 = \lambda_3 = 0 \quad \text{故线性无关}$$

(2) $(\beta_1 \ \beta_2 \ \beta_3) = (\alpha_1 \ \alpha_2 \ \alpha_3) \begin{pmatrix} 2 & 3 & 1 \\ 2 & 1 & 2 \\ 4 & 3 & 3 \end{pmatrix} \Rightarrow \begin{vmatrix} 2 & 3 & 1 \\ 2 & 1 & 2 \\ 4 & 3 & 3 \end{vmatrix} = 2 \neq 0$

$$\Rightarrow \text{rank}(\beta_1, \beta_2, \beta_3) = \text{rank}(\alpha_1, \alpha_2, \alpha_3) = 3$$

故线性无关

9. 设
$$\begin{cases} \beta_1 = \alpha_2 + \alpha_3 + \cdots + \alpha_n, \\ \beta_2 = \alpha_1 + \alpha_3 + \cdots + \alpha_n, \\ \vdots \\ \beta_n = \alpha_1 + \alpha_2 + \cdots + \alpha_{n-1}, \end{cases}$$
 证明向量组 $\alpha_1, \alpha_2, \dots, \alpha_n$ 与向量组 $\beta_1, \beta_2, \dots, \beta_n$ 等价.

证明: 由题可知向量组 $\beta_1, \beta_2, \dots, \beta_n$ 可由向量组 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性表出.

且
$$\beta_1 + \beta_2 + \cdots + \beta_n = (n-1)(\alpha_1 + \alpha_2 + \cdots + \alpha_n)$$

$$\Rightarrow \alpha_1 = \frac{1}{n-1} \sum_{i=1}^n \beta_i - \beta_1$$

$$\alpha_2 = \frac{1}{n-1} \sum_{i=1}^n \beta_i - \beta_2$$

$\dots$

$$\alpha_n = \frac{1}{n-1} \sum_{i=1}^n \beta_i - \beta_n$$

即向量组 $\alpha_1, \alpha_2, \dots, \alpha_n$ 可由
向量组 $\beta_1, \beta_2, \dots, \beta_n$ 线性表出
 $\therefore$ 等价

10. 设向量组 $B: b_1, b_2, \dots, b_r$ 能由向量组 $A: a_1, a_2, \dots, a_s$ 线性表示为

$$(b_1, b_2, \dots, b_r) = (a_1, a_2, \dots, a_s) K,$$

其中 K 为 $s \times r$ 矩阵, 且向量组 A 线性无关, 证明向量组 B 线性无关的充分必要条件是矩阵 K 的秩 $\text{rank}(K) = r$.

证明:

必要性: 向量组 B 线性无关 $\Rightarrow r = \text{rank}(b_1, b_2, \dots, b_r) \leq \text{rank}(K) \leq r$

$\therefore \text{rank}(K) = r$

充分性: 设 $\lambda_1 b_1 + \lambda_2 b_2 + \cdots + \lambda_r b_r = 0$

即 $(b_1, b_2, \dots, b_r) \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_r \end{pmatrix} = 0$

$\Rightarrow (a_1, a_2, \dots, a_s) K \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_r \end{pmatrix} = 0$

由于向量组 A 线性无关 $\Rightarrow K \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_r \end{pmatrix} = 0$

又由于 $\text{rank}(K) = r \Rightarrow \lambda_1 = \lambda_2 = \cdots = \lambda_r = 0$

$\therefore$ 向量组 B 线性无关